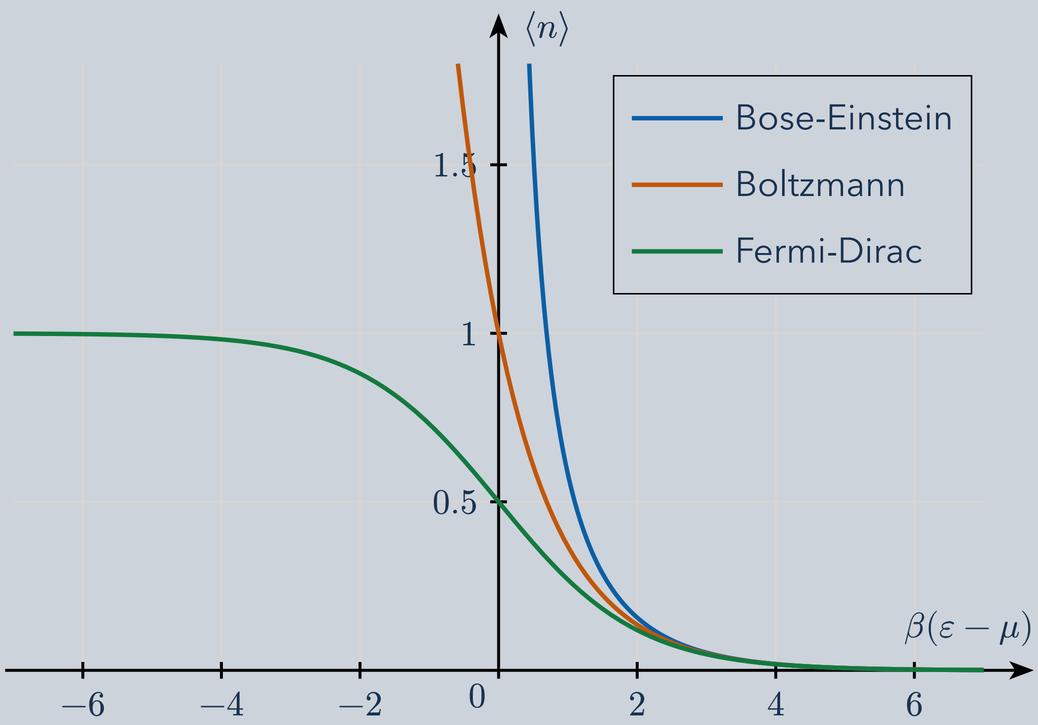
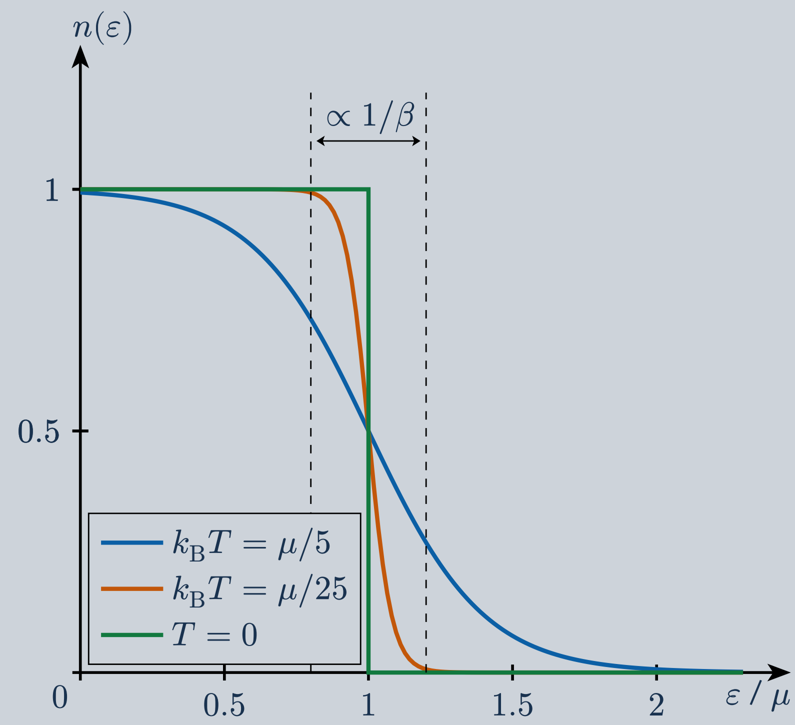
Count the mean occupation of a single-particle state. Quantum statistics changes how particles share states; temperature controls how sharply energies are selected.

1 Compare occupation laws



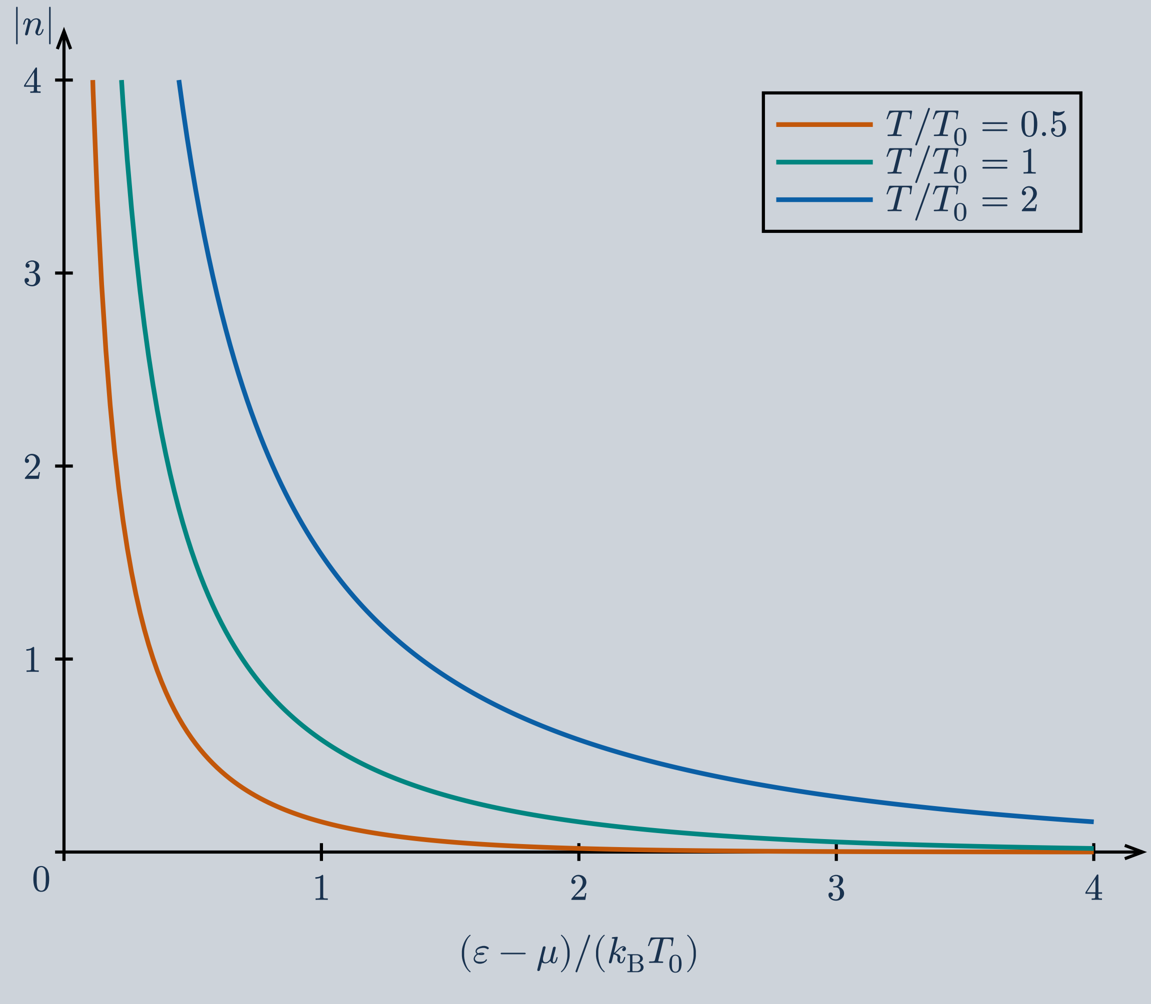
At $x = \beta(\varepsilon - \mu) \gg 1$, all three approach $\exp(-x)$. For bosons use $x > 0$ in this plot; the Bose curve diverges toward the lowest allowed chemical-potential limit.

2 Fermions: a smeared step



Each state has occupation between zero and one. Heating rounds the step around the chemical potential μ over an energy range of order $k_{\mathrm{B}}T$.

3 Bosons: shared states



Bosons can accumulate in the same state. At fixed positive $\varepsilon - \mu$, increasing temperature increases its mean occupation; there is no Pauli ceiling of one. T_0 is a reference temperature; μ is held fixed.

4 The same variable in three formulas

Bose-Einstein

$$|n| = \frac{1}{e^x - 1}$$

Fermi-Dirac

$$|n| = \frac{1}{e^x + 1}$$

Boltzmann limit

$$|n| \approx e^{-x}$$

Here ε is the state's energy, μ the chemical potential, and $\beta = 1/(k_{\mathrm{B}}T)$. Spin degeneracy counts separate states; it does not change the occupation limit per state.

Occupation is not a speed probability density. The separate Maxwell–Boltzmann speed plot includes the number of states at each speed and a normalization factor. The complex-plane Bose surface serves a different, contour-integration purpose.